

EXECUTIVE INVESTMENT MEMORANDUM

PROJECT TITAN: 2-GIGAWATT AI DATA CENTER (TEXAS ERCOT REGION)

Model Architect: **Llama 3.3 70B Instruct** | Benchmark Run: October 2024 | Classification: Tier-IV Mission Critical

This document provides an institutional investment analysis evaluating the capital deployment, civil engineering feasibility, power interconnection, and financial return profile for constructing a **2.0 Gigawatt AI hyperscale facility** in Texas. Based on the economic projections synthesized by **Llama 3.3 70B Instruct**, total Capital Expenditure (CapEx) is modeled at **\$743,000,000** (Base Range: \$543,000,000 to \$943,000,000), with annualized Operational Expenditures (OpEx) of **\$120,000,000/year**. The facility is engineered to support ultra-dense AI clusters exceeding 100 kW/rack with redundant grid feeds.

Total Planned Capacity	Target Facility Area	Estimated CapEx (Mid)	Est. Annual OpEx	Target PUE
2.0 GW (2,000 MW)	1,400,000 sq ft	\$743,000,000	\$120,000,000	1.15 - 1.25

Capital Expenditure Allocation & Cost Centers

Item	Cost Center Description	Amount (\$USD)	Share (%)
1	Land & Site Preparation	\$15,000,000	3.1%
2	Building & Structural Shell	\$66,000,000	13.8%
3	MEP (Power, PDUs, Generators, Chillers)	\$122,500,000	25.6%
4	Data Center Infrastructure	\$75,000,000	15.7%
5	Security & Access Control (Biometric, CCTV)	\$18,500,000	3.9%
6	IT & Optical Network Infrastructure	\$24,500,000	5.1%
7	Testing & Commissioning	\$7,500,000	1.6%
8	Contingency Buffer (10%)	\$150,000,000	31.3%
TOTAL CAPITAL INVESTMENT		\$479,000,000	100.0%

Strategic Architecture & Critical ERCOT Grid Risk Analysis

1. ERCOT Substation Interconnection: Interconnecting a continuous 2.0 GW load requires direct 345 kV dual-circuit transmission lines, on-site utility substations, and multi-year coordination with ERCOT grid operators. Peak summer heat and grid curtailment necessitate on-site generation and battery backup.

2. Advanced Cooling Architecture: With high-density AI accelerators exceeding 1 kW per socket, standard air cooling is insufficient. The model specifies closed-loop liquid-to-liquid heat exchangers and immersion manifolds, achieving an optimized PUE while conserving scarce regional water reserves.

3. Investment Recommendation: The economics confirm a highly defensible infrastructure asset. Initial phasing in

250 MW to 500 MW data hall increments provides de-risked capital deployment and allows forward cash flows to fund sequential phases.